

$$\operatorname{Im}(\langle u, v \rangle) = \frac{\|u - iv\|^2 - \|u + iv\|^2}{4} \quad i \cdot (u - \bar{v}) = -2 \operatorname{Im}(v) \Rightarrow i \cdot (\bar{v} - v) = 2 \operatorname{Im}(v)$$

$$\|u - iv\|^2 = \langle u - iv, u - iv \rangle \stackrel{\text{SL2}}{=} \langle u - iv, u \rangle + \langle u - iv, -iv \rangle$$

$$\stackrel{(1)}{=} \langle u, u \rangle + \langle -iv, u \rangle + \langle u, -iv \rangle + \langle -iv, -iv \rangle$$

$$\stackrel{(2)}{\stackrel{\text{SL7}}{=}} \langle u, u \rangle + i \langle v, u \rangle - i \langle u, v \rangle + \langle v, v \rangle$$

$$\stackrel{\text{SSS}}{=} \langle u, u \rangle + \langle v, v \rangle + i \overline{\langle u, v \rangle} - i \langle u, v \rangle = \|u\|^2 + \|v\|^2 + i(\overline{\langle u, v \rangle} - \langle u, v \rangle)$$

$$\|u + iv\|^2 = \langle u + iv, u + iv \rangle$$

$$\stackrel{\text{SL2}}{=} \langle u + iv, u \rangle + \langle u + iv, iv \rangle$$

$$\stackrel{(3)}{=} \langle u, u \rangle + \langle iv, u \rangle + \langle u, iv \rangle + \langle iv, iv \rangle$$

$$\stackrel{(2)}{=} \langle u, u \rangle - i \langle v, u \rangle + i \langle u, v \rangle + \langle v, v \rangle$$

$$\stackrel{\text{SSS}}{=} \langle u, u \rangle - i \overline{\langle u, v \rangle} + i \langle u, v \rangle + \langle v, v \rangle$$

$$= \langle u, u \rangle + i(\langle u, v \rangle - \overline{\langle u, v \rangle}) + \langle v, v \rangle$$

$$= \|u\|^2 + \|v\|^2 - 2 \operatorname{Im}(\langle u, v \rangle)$$

$$a + bi - a + bi = 2bi = i \cdot 2 \operatorname{Im}(v)$$

$$i \cdot 2bi = -2b = -2 \operatorname{Im}(v)$$

$$v = a + bi \quad \bar{v} = a - bi$$